

Week 5 & 6

Study and Implementation of Git Version Control System

1.Aim

To understand and perform basic operations of Git version control system including repository creation, staging, committing, branching, and pushing code to remote repository.

2.Tools Required

- Git – Distributed version control system
- GitHub – Remote repository hosting service
- Command Line / Git Bash – To execute Git commands

3.Theory

Git is a distributed version control system (DVCS) used to track changes in source code during software development. It allows multiple developers to work simultaneously without overwriting each other's work.

Key Concepts:

- Repository (Repo): Storage location where project files and history are kept
- Version Control: System that records file changes over time
- Commit: Snapshot of changes made in the project
- Branch: Parallel version of the project used for feature development
- Merge: Combining changes from different branches
- Remote Repository: Online repo (GitHub) used for backup and collaboration

4.Architecture Diagram

Working Directory → Staging Area → Local Repository → Remote Repository (GitHub)

5.Procedure

Step 1: Install Git

```
sudo apt install git -y
```

```
git --version
```

Step 2: Configure Git Username and Email

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```

Step 3: Create a New Repository

```
mkdir myproject
```

```
cd myproject  
git init
```

Step 4: Add Files to Repository

```
touch file1.txt  
git add file1.txt
```

Step 5: Commit Changes

```
git commit -m "Initial commit"
```

Step 6: Check Status

```
git status
```

Step 7: Create a Branch

```
git branch feature1  
git checkout feature1
```

Step 8: Merge Branch

```
git checkout main  
git merge feature1
```

Step 9: Connect to GitHub Repository

```
git remote add origin https://github.com/username/repo-name.git
```

Step 10: Push Code to GitHub

```
git push -u origin main
```

6.Common Git Commands

- `git init` – Initialize repository
- `git add .` – Add all files to staging
- `git commit -m "msg"` – Save changes
- `git log` – View commit history
- `git branch` – List branches
- `git merge` – Merge branches
- `git clone <url>` – Copy remote repo
- `git pull` – Fetch and merge updates

7.Result

Successfully performed Git operations such as repository initialization, staging, committing, branching, merging, and pushing code to GitHub.

8.Conclusion

Git version control system helps track code changes, manage multiple versions, and collaborate efficiently. It is an essential tool in modern DevOps and software development practices.